

Roll No.

National Institute of Technology, Kurukshetra
Mid Term-I BTech Examination
September 2025

Course Name: Machine Learning and Data Analytics
Course Code: CSIC-221
Branch: Mechanical Engineering

Semester: 3rd
Max Marks: 15
Time: 50 Minutes

NOTE:

1. Read all questions carefully and attempt each question.
2. Write answers in clear handwritings and provide all calculation steps. Shortcut in calculation part is NOT allowed.

1. Write two common statistical measures to determine shape of the data. Give mathematical definition of each. [1+1]
2. Write down all basic steps of k-means clustering. [2]
3. Explain misclassification rate. What are the cons of minimizing misclassification rate? [1+2]
4. Find $P(2.7)$ by Lagrange's interpolation for the following data: [3]

i	1	2	3
x	1	2	3
y	2.5013	8.6537	11.0821

5. Give mathematical definition of Beta distribution. Compute moments of Beta distribution, where $\alpha = 9$ and $\beta = 7$. [1+4]